Reductions in Cardiovascular Disease Projected from Modest Reductions in Dietary Salt

Background The US diet is high in salt, with the majority coming from processed foods. Reducing dietary salt is an important potential public health target. Methods We used the Coronary Heart Disease (CHD) Policy Model to quantify the benefits of potentially achievable population-wide reductions in dietary salt of up to 3 gm/day (1200 mg/day of sodium). We estimated cardiovascular disease rates and costs in age, sex, and race subgroups, compared salt reduction with other interventions to reduce cardiovascular risk, and determined the cost-effectiveness of salt reduction compared with drug treatment of hypertension. Results Reducing salt by 3 gm/day is projected to result in 60,000–120,000 fewer new CHD cases, 32,000–66,000 fewer new strokes, 54,000–99,000 fewer myocardial infarctions, and 44,000–92,000 fewer deaths from any cause annually. All segments of the population would benefit, with blacks benefiting proportionately more, women benefiting particularly from stroke reduction, older adults from reductions in CHD events, and younger adults from lower mortality rates. The cardiovascular benefits from lower salt are on par with benefits from reducing tobacco, obesity, or cholesterol. A regulatory intervention designed to achieve 3 gm/day salt reduction would save 194,000–392,000 quality-adjusted life-years and $10–24 billion in healthcare costs annually. Such an intervention would be cost-saving even if only a modest 1 gm/day reduction were achieved gradually over the decade from 2010–2019 and would be more cost-effective than treating all hypertensive individuals with medications. Conclusions Modest reduction in dietary salt could substantially reduce cardiovascular events and medical costs and should be a public health target.